

Name: _____

1.1 Basics of Geometry — use the top diagram.

1. Name two lines.

Answer: Sample: line AB, line DC (also line EF)

2. Name two rays with endpoint B.

Answer: Sample: ray BA, ray BE

3. Name three collinear points.

Answer: D, B, and C

1.2 Measuring & Constructing Segments — use the number line.

4. Find PQ and QR.

Answer: $PQ = 6.5$, $QR = 2.5$

5. K is between J and L . $JK = 15$ and $KL = 9$. Find JL .

Answer: $JL = 15 + 9 = 24$

6. S is between R and T . $RS = 3x + 1$, $ST = x - 3$, $RT = 42$. Find RS .

Answer: $4x - 2 = 42 \rightarrow x = 11$; $RS = 34$

1.3 Midpoint & Distance

7. Find the midpoint of $\overline{AB}$ with $A(-8, 5)$ and $B(2, -3)$.

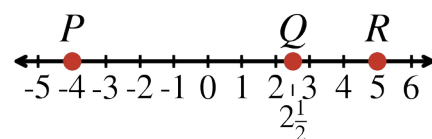
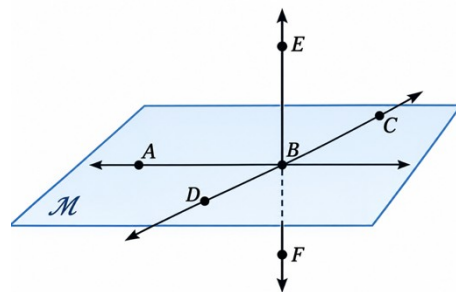
Answer: $(-3, 1)$

8. $M(-1, 2)$ is the midpoint of $\overline{AB}$, and A has coordinates $(4, 6)$. Find B .

Answer: $B = (-6, -2)$

9. Find the distance between $(-2, 3)$ and $(4, -5)$.

Answer: $\sqrt{6^2 + 8^2} = 10$



1.4 Perimeter & Area

- 10.** What do you call a polygon with 7 sides? Is a polygon with a "dent" convex or concave?

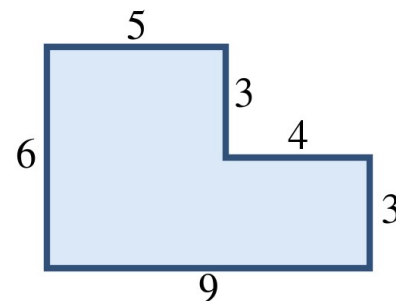
Answer: Heptagon; concave

- 11.** A rectangle has vertices $(-5, 2)$, $(3, 2)$, $(3, -3)$, and $(-5, -3)$. Find its perimeter and area.

Answer: Length 8, width 5 $\rightarrow P = 26$ units, $A = 40$ square units

- 12.** Find the perimeter and area of the shaded figure.

Answer: $P = 9 + 3 + 4 + 3 + 5 + 6 = 30$ units; $A = 9 \cdot 3 + 5 \cdot 3 = 42$ square units



1.5 Measuring & Constructing Angles

- 13.** $m\angle ABD = 27^\circ$ and $m\angle DBC = 44^\circ$, with D in the interior of $\angle ABC$. Find $m\angle ABC$.

Answer: $27^\circ + 44^\circ = 71^\circ$

- 14.** $\angle XYZ$ is a straight angle divided into angles measuring $(9x + 11)^\circ$ and $(5x + 1)^\circ$. Find both measures.

Answer: $14x + 12 = 180 \rightarrow x = 12$; 119° and 61°

- 15.** Ray WY bisects $\angle XWZ$. $m\angle XWY = (6x + 2)^\circ$ and $m\angle YWZ = (8x - 12)^\circ$. Find $m\angle XWZ$.

Answer: $6x + 2 = 8x - 12 \rightarrow x = 7$; each half is 44° , so $m\angle XWZ = 88^\circ$

1.6 Describing Pairs of Angles — use the bottom diagram for Exercise 7.

- 16.** In the diagram, find $m\angle 1$, $m\angle 2$, and $m\angle 3$.

Answer: $m\angle 2 = 64^\circ$ (vertical); $m\angle 1 = m\angle 3 = 116^\circ$ (linear pairs)

- 17.** Find the complement and the supplement of a 71° angle.

Answer: Complement 19° ; supplement 109°

- 18.** Two angles that form a linear pair measure $(5x + 12)^\circ$ and $(3x + 8)^\circ$. Find both measures.

Answer: $8x + 20 = 180 \rightarrow x = 20$; 112° and 68°

